Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 14: Two-Port Circuits

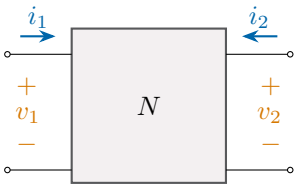
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Colour code used in these notes and in the slides

■ current i ■ voltage v, node voltage e

Motivation. Previously, a one-port without independent sources was described by one number, its input resistance: $v = R_{\text{in}}\, i$. Many circuits have an input and an output: an amplifier, a transformer, a filter. Such a circuit has two ports, and its description is the natural generalization of $v = R\, i$: two port voltages, two port currents and a 2×2 matrix. This lecture introduces four such matrices (**resistance**, **conductance**, **hybrid** and **transmission**) and the **power** of a two-port.

1. Two-port circuits



N is an LTI resistive circuit with no independent sources. At each port the current enters at the + terminal and leaves at the − terminal. There are four port variables v_1, i_1, v_2, i_2 and two equations between them: two of the variables can be written in terms of the other two.

given	find	matrix	equation
i_1, i_2	v_1, v_2	resistance $\mathbf{R}$	$[\, v_1 \;\; v_2 \,]^T = \mathbf{R}\, [\, i_1 \;\; i_2 \,]^T$
v_1, v_2	i_1, i_2	conductance $\mathbf{G}$	$[\, i_1 \;\; i_2 \,]^T = \mathbf{G}\, [\, v_1 \;\; v_2 \,]^T$
i_1, v_2	v_1, i_2	hybrid $\mathbf{H}$	$[\, v_1 \;\; i_2 \,]^T = \mathbf{H}\, [\, i_1 \;\; v_2 \,]^T$
v_1, i_2	i_1, v_2	hybrid $\mathbf{H}'$	$[\, i_1 \;\; v_2 \,]^T = \mathbf{H}'\, [\, v_1 \;\; i_2 \,]^T$
v_2, $-i_2$	v_1, i_1	transmission $\mathbf{T}$	$[\, v_1 \;\; i_1 \,]^T = \mathbf{T}\, [\, v_2 \;\; -i_2 \,]^T$

2. The resistance matrix

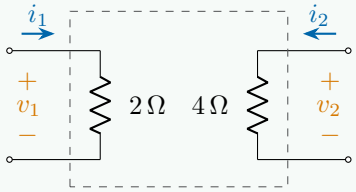
Resistance matrix

The port voltages in terms of the port currents:

$$v_1 = r_{11}\, i_1 + r_{12}\, i_2$$
$$v_2 = r_{21}\, i_1 + r_{22}\, i_2$$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \underbrace{\begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix}}_{\mathbf{R}} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

Example 14.1 — two separate resistors



The two-port is a $2\,\Omega$ resistor across port 1 and a $4\,\Omega$ resistor across port 2. Find $\mathbf{R}$.

Solution

Step 1. Each port alone

Each resistor carries only its own port current:

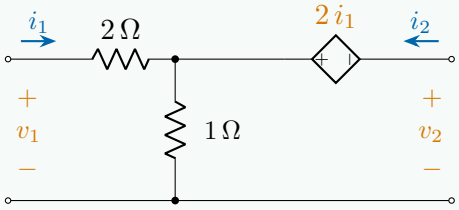
$$v_1 = 2\,i_1 + 0 \cdot i_2, \qquad v_2 = 0 \cdot i_1 + 4\,i_2.$$

Step 2. The matrix

$$\mathbf{R} = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \Omega.$$

The off-diagonal entries are zero: the two ports are not coupled.

Example 14.2 — a dependent source inside



The two-port contains a $2\,\Omega$ and a $1\,\Omega$ resistor and a current-controlled voltage source $2\,i_1$. Find $\mathbf{R}$.

Solution

Step 1. The current of $1\,\Omega$

By KCL at the middle node, the $1\,\Omega$ resistor carries $i_1 + i_2$ downwards.

Step 2. KVL from each port

From port 1 through $2\,\Omega$ and $1\,\Omega$:

$$v_1 = 2\,i_1 + (i_1 + i_2) = 3\,i_1 + i_2.$$

From port 2 through the dependent source and $1\,\Omega$:

$$v_2 = -2\,i_1 + (i_1 + i_2) = -i_1 + i_2.$$

Step 3. The matrix

Collect the terms in i_1 and i_2:

$$\mathbf{R} = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix} \Omega.$$

Without the dependent source $\mathbf{R}$ would be symmetric; here $r_{12} \neq r_{21}$.

2.1. The entries from open-circuit tests

The matrix equation in the box above also tells how to find $\mathbf{R}$ one column at a time: make one port current zero, that is, leave that port **open**.

Port 2 open ($i_2 = 0$). Drive port 1 with a current i_1, for example with a current source. The second column of $\mathbf{R}$ is multiplied by zero and drops out:

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix} \begin{bmatrix} i_1 \\ 0 \end{bmatrix} = \begin{bmatrix} r_{11}\,i_1 \\ r_{21}\,i_1 \end{bmatrix}.$$

Both port voltages are proportional to i_1. Dividing by i_1 gives the **first column**:

$$r_{11} = \left.\frac{v_1}{i_1}\right|_{i_2=0}, \qquad r_{21} = \left.\frac{v_2}{i_1}\right|_{i_2=0}.$$

Port 1 open ($i_1 = 0$). Drive port 2 with a current i_2. Now the first column drops out:

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix} \begin{bmatrix} 0 \\ i_2 \end{bmatrix} = \begin{bmatrix} r_{12}\,i_2 \\ r_{22}\,i_2 \end{bmatrix},$$

and dividing by i_2 gives the **second column**:

$$r_{12} = \left.\frac{v_1}{i_2}\right|_{i_1=0}, \qquad r_{22} = \left.\frac{v_2}{i_2}\right|_{i_1=0}.$$

What the entries mean. The diagonal entries are input resistances: r_{11} is the resistance seen at port 1 when port 2 is open, and r_{22} is the resistance seen at port 2 when port 1 is open. The off-diagonal entries are transfer resistances: r_{21} is the voltage that appears at the open port 2 per ampere into port 1, and r_{12} the other

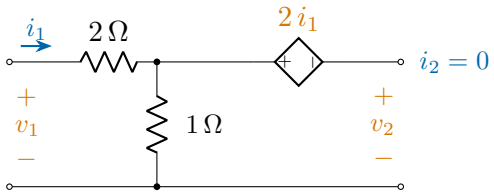
way round. Since both tests leave a port open, **R** is also called the **open-circuit resistance matrix**. Each test is an ordinary one-source linear circuit.

Example 14.2 revisited — open-circuit tests

Find **R** of Example 14.2 again, this time from the two open-circuit tests.

Solution

Step 1. Port 2 open: $i_2 = 0$

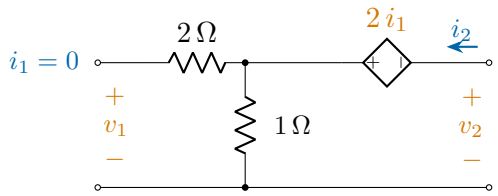


No current flows through the dependent source, so v_2 is the voltage of 1 Ω minus the dependent source:

$$v_1 = (2+1)\, i_1 = 3\, i_1$$
$$v_2 = i_1 - 2\, i_1 = -i_1$$

So $r_{11} = 3\,\Omega$ and $r_{21} = -1\,\Omega$.

Step 2. Port 1 open: $i_1 = 0$



No current flows in 2 Ω, and the dependent source is $2\, i_1 = 0$:

$$v_1 = i_2$$
$$v_2 = i_2$$

So $r_{12} = 1\,\Omega$ and $r_{22} = 1\,\Omega$: the same **R** as before. ✓

3. The conductance matrix

Conductance matrix

The port currents in terms of the port voltages:

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \underbrace{\begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}}_{\mathbf{G}} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}.$$

The entries follow from **short-circuit tests**:

$$g_{11} = \left.\frac{i_1}{v_1}\right|_{v_2=0}, \quad g_{21} = \left.\frac{i_2}{v_1}\right|_{v_2=0},$$
$$g_{12} = \left.\frac{i_1}{v_2}\right|_{v_1=0}, \quad g_{22} = \left.\frac{i_2}{v_2}\right|_{v_1=0}.$$

The entries, column by column. Now make one port *voltage* zero, that is, **short** that port, and drive the other port with a voltage.

Port 2 shorted ($v_2 = 0$):

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ 0 \end{bmatrix} = \begin{bmatrix} g_{11}\, v_1 \\ g_{21}\, v_1 \end{bmatrix}$$

the **first column**.

Port 1 shorted ($v_1 = 0$):

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} 0 \\ v_2 \end{bmatrix} = \begin{bmatrix} g_{12}\, v_2 \\ g_{22}\, v_2 \end{bmatrix}$$

the **second column**.

What the entries mean. The diagonal entries are input conductances: g_{11} is the conductance seen at port 1 when port 2 is shorted, and g_{22} the conductance seen at port 2 when port 1 is shorted. The off-diagonal entries are transfer conductances: g_{21} is the current that flows in the shorted port 2 per volt at port 1, and g_{12} the other way round. Since both tests short a port, $\mathbf{G}$ is also called the **short-circuit conductance matrix**.

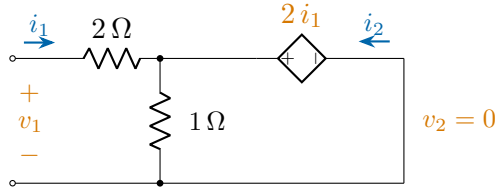
If $\mathbf{R}$ is invertible, $\mathbf{v} = \mathbf{R}\,\mathbf{i}$ gives $\mathbf{i} = \mathbf{R}^{-1}\mathbf{v}$, so $\mathbf{G} = \mathbf{R}^{-1}$ (and $\mathbf{R} = \mathbf{G}^{-1}$). Note that $g_{11} \neq 1/r_{11}$ in general: the two are measured with the other port shorted and open, respectively.

Example 14.3 — the conductance matrix of Example 14.2

Find the first column of $\mathbf{G}$ for the two-port of Example 14.2 by a short-circuit test. Then find all of $\mathbf{G}$.

Solution

Step 1. Port 2 shorted: $v_2 = 0$



Two meshes, with i_1 and i_2 as the mesh currents.

$$\text{Mesh 1:} \quad -v_1 + 2\,i_1 + (i_1 + i_2) = 0$$
$$\text{Mesh 2:} \quad -2\,i_1 + (i_1 + i_2) = 0$$

Step 2. Solve

Mesh 2 gives $i_2 = i_1$. Mesh 1 then gives $v_1 = 2\,i_1 + 2\,i_1 = 4\,i_1$.

Step 3. The first column

$$g_{11} = \left.\frac{i_1}{v_1}\right|_{v_2=0} = \frac{1}{4}\,\text{S}, \qquad g_{21} = \left.\frac{i_2}{v_1}\right|_{v_2=0} = \frac{1}{4}\,\text{S}.$$

Step 4. All of G

The second test ($v_1 = 0$) gives the second column in the same way. The result is the inverse of $\mathbf{R}$:

$$\mathbf{G} = \mathbf{R}^{-1} = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}^{-1} = \frac{1}{4}\begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & -\frac{1}{4} \\ \frac{1}{4} & \frac{3}{4} \end{bmatrix} \text{S}.$$

Check: $\mathbf{G}\,\mathbf{R} = \frac{1}{4}\begin{bmatrix} 3+1 & 1-1 \\ 3-3 & 1+3 \end{bmatrix} = \mathbf{1}$. ✓

Note that $g_{11} = \frac{1}{4}\,\text{S} \neq 1/r_{11} = \frac{1}{3}\,\text{S}$.

Remark

$\mathbf{R}$ or $\mathbf{G}$ need not exist. When both exist, $\mathbf{G} = \mathbf{R}^{-1}$. The ideal transformer (Example 14.5) has neither. Section 6 explains when a matrix exists.

4. Hybrid matrices

Hybrid matrices

Mix one current and one voltage on each side:

$$\begin{bmatrix} v_1 \\ i_2 \end{bmatrix} = \underbrace{\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}}_{\mathbf{H}} \begin{bmatrix} i_1 \\ v_2 \end{bmatrix}, \qquad \begin{bmatrix} i_1 \\ v_2 \end{bmatrix} = \underbrace{\begin{bmatrix} h'_{11} & h'_{12} \\ h'_{21} & h'_{22} \end{bmatrix}}_{\mathbf{H}'} \begin{bmatrix} v_1 \\ i_2 \end{bmatrix},$$

with $\mathbf{H}' = \mathbf{H}^{-1}$ when the inverse exists.

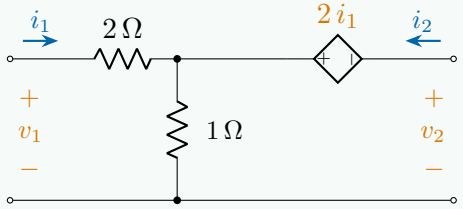
The entries of $\mathbf{H}$ follow from two tests: port 2 shorted ($v_2 = 0$) gives $h_{11} = v_1/i_1$ and $h_{21} = i_2/i_1$; port 1 open ($i_1 = 0$) gives $h_{12} = v_1/v_2$ and $h_{22} = i_2/v_2$. The entries have different units: h_{11} is a resistance, h_{22} a conductance, h_{12} and h_{21} have no units.

The second hybrid matrix $\mathbf{H}'$ has the roles of the two ports swapped:

$$\begin{aligned} i_1 &= h'_{11}\,v_1 + h'_{12}\,i_2 \\ v_2 &= h'_{21}\,v_1 + h'_{22}\,i_2 \end{aligned} \qquad \begin{bmatrix} i_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} h'_{11} & h'_{12} \\ h'_{21} & h'_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ i_2 \end{bmatrix}.$$

Its inputs are v_1 and i_2, so its tests are the other way round: port 2 open ($i_2 = 0$) gives $h'_{11} = i_1/v_1$ and $h'_{21} = v_2/v_1$; port 1 shorted ($v_1 = 0$) gives $h'_{12} = i_1/i_2$ and $h'_{22} = v_2/i_2$. Now h'_{11} is a conductance, h'_{22} a resistance, and h'_{12}, h'_{21} have no units. When both exist, $\mathbf{H}' = \mathbf{H}^{-1}$.

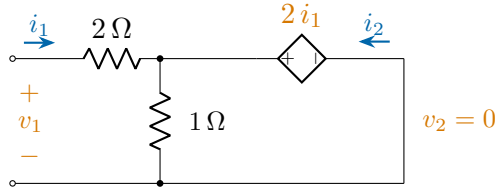
Example 14.4 — the hybrid matrix of a circuit



This is the two-port of Example 14.2. Find **H**.

Solution

Step 1. Port 2 shorted: $v_2 = 0$



Mesh 2 contains the dependent source and 1 Ω:

$$-2\,i_1 + (i_1 + i_2) = 0$$

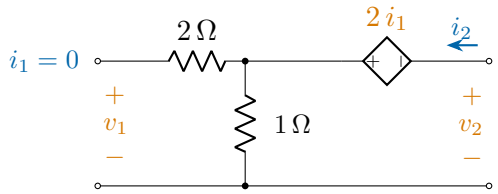
$$\Rightarrow \; i_2 = i_1.$$

Mesh 1 then gives

$$v_1 = 2\,i_1 + (i_1 + i_2) = 4\,i_1.$$

So $h_{11} = 4\,\Omega$ and $h_{21} = 1$.

Step 2. Port 1 open: $i_1 = 0$



The dependent source is $2\,i_1 = 0$, a short circuit, and no current flows in 2 Ω:

$$v_2 = 1 \cdot i_2 \;\Rightarrow\; i_2 = v_2.$$

The voltage of port 1 is the voltage of 1 Ω:

$$v_1 = i_2 = v_2.$$

So $h_{12} = 1$ and $h_{22} = 1\,\mathrm{S}$.

Step 3. The matrix

$$\mathbf{H} = \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix}.$$

Check: Example 14.2 gives $\mathbf{R} = \left[\begin{smallmatrix} 3 & 1 \\ -1 & 1 \end{smallmatrix}\right]$. Solving its second row for i_2 and substituting into the first row gives the same **H**. ✓

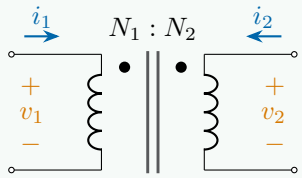
Step 4. The other hybrid matrix

$\det \mathbf{H} = 4 - 1 = 3$, so

$$\mathbf{H}' = \mathbf{H}^{-1} = \frac{1}{3}\begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{4}{3} \end{bmatrix}.$$

Check with the tests of $\mathbf{H}'$. Port 2 open: $v_1 = 3\,i_1$ and $v_2 = -i_1$, so $h'_{11} = \frac{1}{3}\,\mathrm{S}$ and $h'_{21} = -\frac{1}{3}$. Port 1 shorted: $3\,i_1 + i_2 = 0$ and $v_2 = -i_1 + i_2 = \frac{4}{3}\,i_2$, so $h'_{12} = -\frac{1}{3}$ and $h'_{22} = \frac{4}{3}\,\Omega$. ✓

Example 14.5 — the hybrid matrix of the ideal transformer



The ideal transformer $N_1 : N_2$ (previously) satisfies

$$\frac{v_1}{N_1} = \frac{v_2}{N_2}, \qquad N_1\,i_1 + N_2\,i_2 = 0.$$

Find $\mathbf{H}$. Does $\mathbf{R}$ exist?

Solution

Step 1. Solve for v_1 and i_2

With the turns ratio $n = N_1/N_2$:

$$v_1 = n\,v_2, \qquad i_2 = -n\,i_1.$$

Step 2. The matrix

$$\begin{bmatrix} v_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 & n \\ -n & 0 \end{bmatrix}\begin{bmatrix} i_1 \\ v_2 \end{bmatrix} \qquad \Rightarrow \qquad \mathbf{H} = \begin{bmatrix} 0 & n \\ -n & 0 \end{bmatrix}.$$

Its inverse is the other hybrid matrix:

$$\mathbf{H}' = \mathbf{H}^{-1} = \begin{bmatrix} 0 & -1/n \\ 1/n & 0 \end{bmatrix}.$$

Step 3. No resistance matrix

With both ports open, $i_1 = i_2 = 0$, but $v_1 = n\,v_2$ can take any value. So the voltages are not determined by the currents: $\mathbf{R}$ does not exist. In the same way, shorting both ports ($v_1 = v_2 = 0$) leaves $i_1 = -i_2/n$ free, so $\mathbf{G}$ does not exist either.

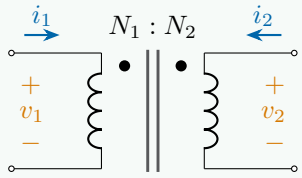
5. The transmission matrix

Transmission matrix (chain parameters)

$$\begin{bmatrix} v_1 \\ i_1 \end{bmatrix} = \underbrace{\begin{bmatrix} A & B \\ C & D \end{bmatrix}}_{\mathbf{T}} \begin{bmatrix} v_2 \\ -i_2 \end{bmatrix}$$

Port 1 in terms of port 2. The minus sign makes $-i_2$ the current that leaves port 2, for example towards a load.

Example 14.6 — the transmission matrix of the ideal transformer



Find $\mathbf{T}$ for the ideal transformer $N_1 : N_2$.

Solution

Step 1. Port 1 in terms of port 2

From the two transformer equations:

$$v_1 = \frac{N_1}{N_2}\, v_2, \qquad i_1 = -\frac{N_2}{N_1}\, i_2 = \frac{N_2}{N_1}\,(-i_2).$$

Step 2. The matrix

$$\begin{bmatrix} v_1 \\ i_1 \end{bmatrix} = \begin{bmatrix} N_1/N_2 & 0 \\ 0 & N_2/N_1 \end{bmatrix} \begin{bmatrix} v_2 \\ -i_2 \end{bmatrix} \quad \Rightarrow \quad \mathbf{T} = \begin{bmatrix} N_1/N_2 & 0 \\ 0 & N_2/N_1 \end{bmatrix}.$$

Which test? Open or short

Every matrix has two *inputs*: the variables in the vector on the right-hand side. To get one column, set the other input to zero. A zero current means that port is **open**; a zero voltage means that port is **shorted**. (This is the same as killing a source: a dead current source is an open, a dead voltage source is a short.)

Matrix	Inputs	1st column	2nd column
$\mathbf{R}$: $[\,v_1\ v_2\,]^T = \mathbf{R}\,[\,i_1\ i_2\,]^T$	i_1, i_2	$i_2 = 0$: port 2 **open**	$i_1 = 0$: port 1 **open**
$\mathbf{G}$: $[\,i_1\ i_2\,]^T = \mathbf{G}\,[\,v_1\ v_2\,]^T$	v_1, v_2	$v_2 = 0$: port 2 **shorted**	$v_1 = 0$: port 1 **shorted**
$\mathbf{H}$: $[\,v_1\ i_2\,]^T = \mathbf{H}\,[\,i_1\ v_2\,]^T$	i_1, v_2	$v_2 = 0$: port 2 **shorted**	$i_1 = 0$: port 1 **open**
$\mathbf{H}'$: $[\,i_1\ v_2\,]^T = \mathbf{H}'\,[\,v_1\ i_2\,]^T$	v_1, i_2	$i_2 = 0$: port 2 **open**	$v_1 = 0$: port 1 **shorted**
$\mathbf{T}$: $[\,v_1\ i_1\,]^T = \mathbf{T}\,[\,v_2\ -i_2\,]^T$	v_2, $-i_2$	$i_2 = 0$: port 2 **open**	$v_2 = 0$: port 2 **shorted**

The names already say it: **R** is the open-circuit matrix and **G** the short-circuit matrix. **H** mixes a current and a voltage as inputs, so its tests are mixed too, and its entries have mixed units. **T** is the odd one: both tests are done at port 2, and port 1 is only measured. The port that is not zeroed is driven by a source of its own input type: a current source if its input is a current, a voltage source if it is a voltage.

6. When does a matrix exist?

Each matrix makes a promise: *choose its two inputs freely, and the two outputs are then fixed.* Some two-ports break this promise for some matrices. Then that matrix does not exist.

6.1. The test: both inputs zero

Set both inputs to zero at the same time.

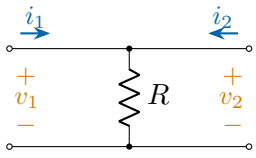
- If the circuit then forces all four port variables to zero, the matrix **exists**.
- If some port variable can still be nonzero, the matrix **does not exist**.

Why: a nonzero solution with zero inputs can be added to any other solution. The inputs stay the same, but the outputs change. So the inputs do not fix the outputs.

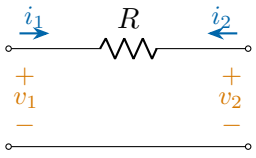
Matrix	Inputs set to zero	In the circuit
R	$i_1 = i_2 = 0$	both ports **open**
G	$v_1 = v_2 = 0$	both ports **shorted**
H	$i_1 = 0$, $v_2 = 0$	port 1 open, port 2 shorted
H′	$v_1 = 0$, $i_2 = 0$	port 1 shorted, port 2 open
T	$v_2 = 0$, $i_2 = 0$	port 2 **open and shorted**: no voltage, no current

For **T** both conditions are at port 2. Port 1 is left alone. If port 1 can still carry a current or a voltage, port 2 cannot tell what happens at port 1, and **T** does not exist.

6.2. Three simple cases



(a) one resistor across both ports



(b) one resistor between the ports

(a) No G. Short both ports. The two shorts and the wires form a loop around the resistor. A current can enter port 1 and leave port 2 through the wires: $i_1 = -i_2$ can be anything. So **G** does not exist. **R** does exist:

$$v_1 = v_2 = R\,(i_1 + i_2) \qquad \Rightarrow \qquad \mathbf{R} = \begin{bmatrix} R & R \\ R & R \end{bmatrix}, \qquad \det \mathbf{R} = 0.$$

$\mathbf{R}$ is not invertible, as it must be: if $\mathbf{G}$ existed, it would be $\mathbf{R}^{-1}$.

(b) No R. Open both ports. Nothing connects the top terminals to the bottom ones, so $v_1 = v_2$ can be any value. So $\mathbf{R}$ does not exist. $\mathbf{G}$ does exist:

$$i_1 = -i_2 = \frac{v_1 - v_2}{R} \qquad \Rightarrow \qquad \mathbf{G} = \frac{1}{R}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad \det \mathbf{G} = 0.$$

(c) No T: two separate resistors (Example 14.1). Make port 2 dead: $v_2 = 0$ and $i_2 = 0$. The $2\,\Omega$ resistor at port 1 does not notice: any i_1 with $v_1 = 2\,i_1$ is still possible. So $\mathbf{T}$ does not exist. In words: port 2 knows nothing about port 1, so port 1 cannot be written in terms of port 2.

The ideal transformer (Example 14.5) has neither $\mathbf{R}$ nor $\mathbf{G}$, but $\mathbf{H}$, $\mathbf{H}'$ and $\mathbf{T}$ exist. Cases (a) and (b) both have $\mathbf{T}$: $\left[\begin{smallmatrix} 1 & 0 \\ 1/R & 1 \end{smallmatrix}\right]$ and $\left[\begin{smallmatrix} 1 & R \\ 0 & 1 \end{smallmatrix}\right]$.

6.3. From one matrix to another

When one matrix is known, the others follow by solving its two equations for the other variables. Each step divides by one number, and that number must not be zero. For example, $\mathbf{T}$ from $\mathbf{R}$: the second row $v_2 = r_{21}\,i_1 + r_{22}\,i_2$ gives

$$i_1 = \frac{1}{r_{21}}\,v_2 + \frac{r_{22}}{r_{21}}\,(-i_2) \qquad (\text{needs } r_{21} \neq 0),$$

and the first row then gives v_1. The result is

$$\mathbf{T} = \frac{1}{r_{21}}\begin{bmatrix} r_{11} & \det \mathbf{R} \\ 1 & r_{22} \end{bmatrix}.$$

The other cases work the same way:

Wanted	**Needs**	**If it fails**
$\mathbf{G} = \mathbf{R}^{-1}$	$\det \mathbf{R} \neq 0$	with both ports shorted a current can still flow (case a)
$\mathbf{H}$ from $\mathbf{R}$	$r_{22} \neq 0$	with port 1 open, port 2 is a short: v_2 cannot be chosen
$\mathbf{H}'$ from $\mathbf{R}$	$r_{11} \neq 0$	with port 2 open, port 1 is a short: v_1 cannot be chosen
$\mathbf{H}' = \mathbf{H}^{-1}$	$\det \mathbf{H} \neq 0$	v_1 and i_2 cannot both be chosen freely
$\mathbf{T}$ from $\mathbf{R}$	$r_{21} \neq 0$	port 1 has no effect on port 2 (case c)
$\mathbf{T}$ from $\mathbf{G}$	$g_{21} \neq 0$	port 1 has no effect on port 2

The formulas:

$$\mathbf{H} = \frac{1}{r_{22}}\begin{bmatrix} \det \mathbf{R} & r_{12} \\ -r_{21} & 1 \end{bmatrix}, \qquad \mathbf{H}' = \frac{1}{r_{11}}\begin{bmatrix} 1 & -r_{12} \\ r_{21} & \det \mathbf{R} \end{bmatrix}, \qquad \mathbf{T} = -\frac{1}{g_{21}}\begin{bmatrix} g_{22} & 1 \\ \det \mathbf{G} & g_{11} \end{bmatrix}.$$

Check with Example 14.2: $\mathbf{R} = \left[\begin{smallmatrix} 3 & 1 \\ -1 & 1 \end{smallmatrix}\right]$, $\det \mathbf{R} = 4$, $r_{22} = 1$, so $\mathbf{H} = \left[\begin{smallmatrix} 4 & 1 \\ 1 & 1 \end{smallmatrix}\right]$, as found before. And $r_{21} = -1 \neq 0$, so $\mathbf{T}$ exists: $\mathbf{T} = -\left[\begin{smallmatrix} 3 & 4 \\ 1 & 1 \end{smallmatrix}\right]$.

7. Measuring a two-port

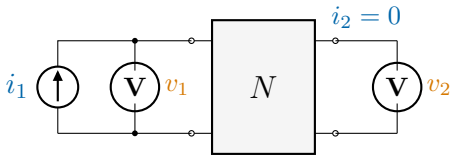
In a laboratory the two-port is a closed box with four terminals. Nothing inside is known, but if it is linear, $\mathbf{R}$ (or $\mathbf{G}$, $\mathbf{H}$, $\mathbf{T}$) has only four numbers. Each experiment gives two equations, so **two experiments** are enough. The open- and short-circuit tests are the two experiments that give the entries most directly.

7.1. Ideal meters are opens and shorts

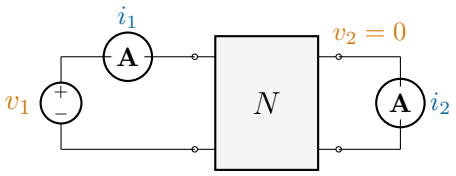
An ideal **voltmeter** draws no current: it is an *open* circuit. An ideal **ammeter** has no voltage across it: it is a *short* circuit. So

- a port with only a voltmeter across it is an **open** port, and the meter reads its voltage;
- a port closed by only an ammeter is a **shorted** port, and the meter reads its current.

The tests of this lecture are therefore real experiments: a source at one port, and meters at both ports.



Open-circuit test. A current source drives port 1; voltmeters read v_1 and v_2. Then $r_{11} = v_1/i_1$ and $r_{21} = v_2/i_1$.



Short-circuit test. A voltage source drives port 1; ammeters read i_1 and i_2. Then $g_{11} = i_1/v_1$ and $g_{21} = i_2/v_1$.

Each entry is one meter reading divided by the source value; no circuit analysis is needed. Moving the source to port 2 gives the second column. For $\mathbf{H}$, port 2 gets an ammeter and port 1 a voltmeter, and so on, following the table above.

7.2. When a port cannot be opened or shorted

Opening or shorting a port is not always possible. A short at the output of an amplifier may damage it. An open-circuit test may be impossible: with port 2 of an ideal transformer open, no current can enter port 1, so its $\mathbf{R}$ does not exist. Then any **two independent experiments** still work. Put the two measured current vectors side by side, and the two voltage vectors side by side:

$$\underbrace{\begin{bmatrix} v_1^{(1)} & v_1^{(2)} \\ v_2^{(1)} & v_2^{(2)} \end{bmatrix}}_{\mathbf{V}} = \mathbf{R} \underbrace{\begin{bmatrix} i_1^{(1)} & i_1^{(2)} \\ i_2^{(1)} & i_2^{(2)} \end{bmatrix}}_{\mathbf{I}} \quad \Rightarrow \quad \mathbf{R} = \mathbf{V}\,\mathbf{I}^{-1}.$$

The superscript is the experiment number. The experiments are independent when $\mathbf{I}$ is invertible. The open-circuit tests are the special case $\mathbf{I}$ diagonal, where no inverse has to be computed.

Example 14.7 — R from two measurements

Two experiments on a linear two-port give

	i_1 (A)	i_2 (A)	v_1 (V)	v_2 (V)
Experiment 1	1	1	5	5
Experiment 2	1	−1	3	−1

Neither experiment has an open port. Find $\mathbf{R}$.

Solution

Step 1. Stack the experiments

$$\begin{bmatrix} 5 & 3 \\ 5 & -1 \end{bmatrix} = \mathbf{R} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Step 2. Solve

$\det \mathbf{I} = -2 \neq 0$, so the experiments are independent, and

$$\mathbf{I}^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad \mathbf{R} = \begin{bmatrix} 5 & 3 \\ 5 & -1 \end{bmatrix} \cdot \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}.$$

Step 3. Check

Experiment 1: $\mathbf{R}\,[1\ 1]^T = [5\ 5]^T$. Experiment 2: $\mathbf{R}\,[1\ -1]^T = [3\ -1]^T$. ✓

8. The power of a two-port

Let P_N be the sum of the powers absorbed by all elements inside the two-port N.

Theorem

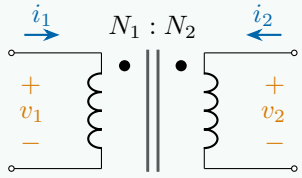
$$P_N = v_1\, i_1 + v_2\, i_2.$$

Why. Connect sources to the two ports. By Tellegen's theorem (previously), the sum of the absorbed powers of all branches is zero. The port sources absorb $-v_1\, i_1$ and $-v_2\, i_2$, so the elements inside N absorb $v_1\, i_1 + v_2\, i_2$.

Passive and active two-ports

A two-port is **passive** if $P_N \geq 0$ for all possible port variables (v_1, i_1, v_2, i_2). It is **active** if it is not passive.

Example 14.8 — is the ideal transformer passive?



Find the power P_N of the ideal transformer. Is it passive or active?

Solution

Step 1. The power

Plug $v_1 = n\, v_2$ and $i_2 = -n\, i_1$ into the theorem:

$$P_N = v_1\, i_1 + v_2\, i_2 = n\, v_2\, i_1 - n\, v_2\, i_1 = 0.$$

Step 2. Passive or active

$P_N = 0 \geq 0$ for all port variables, so the ideal transformer is passive. It neither absorbs nor delivers power: what enters one port leaves the other.

Summary

Matrix	Equation	How to find it
resistance $\mathbf{R}$	$[\, v_1 \;\; v_2 \,]^T = \mathbf{R}\, [\, i_1 \;\; i_2 \,]^T$	KVL, or open-circuit tests ($i_2 = 0$, then $i_1 = 0$)
conductance $\mathbf{G}$	$[\, i_1 \;\; i_2 \,]^T = \mathbf{G}\, [\, v_1 \;\; v_2 \,]^T$	short-circuit tests; $\mathbf{G} = \mathbf{R}^{-1}$ when both exist
hybrid $\mathbf{H}$, $\mathbf{H}'$	$[\, v_1 \;\; i_2 \,]^T = \mathbf{H}\, [\, i_1 \;\; v_2 \,]^T$	port 2 shorted, then port 1 open; $\mathbf{H}' = \mathbf{H}^{-1}$
transmission $\mathbf{T}$	$[\, v_1 \;\; i_1 \,]^T = \mathbf{T}\, [\, v_2 \;\; -i_2 \,]^T$	port 1 in terms of port 2
existence	set both inputs to zero	exists if all port variables are then forced to zero
power	$P_N = v_1\, i_1 + v_2\, i_2$	passive if $P_N \geq 0$ for all port variables

A. Passivity from the matrix

A two-port is passive if $P_N \geq 0$ for all port variables. When $\mathbf{R}$, $\mathbf{G}$ or $\mathbf{H}$ exists, this can be decided from the matrix alone.

A.1. Power as a quadratic form

Write $\mathbf{i} = [\,i_1 \;\; i_2\,]^T$ and $\mathbf{v} = [\,v_1 \;\; v_2\,]^T$. With $\mathbf{v} = \mathbf{R}\,\mathbf{i}$,

$$P_N = v_1\, i_1 + v_2\, i_2 = \mathbf{i}^T\mathbf{v} = \mathbf{i}^T\mathbf{R}\,\mathbf{i} = r_{11}\, {i_1}^2 + (r_{12} + r_{21})\, i_1\, i_2 + r_{22}\, {i_2}^2.$$

Only the sum $r_{12} + r_{21}$ appears. So only the **symmetric part** of $\mathbf{R}$ matters:

$$\mathbf{R}_s = \tfrac{1}{2}\,(\mathbf{R} + \mathbf{R}^T) = \begin{bmatrix} r_{11} & \tfrac{1}{2}(r_{12} + r_{21}) \\ \tfrac{1}{2}(r_{12} + r_{21}) & r_{22} \end{bmatrix}, \qquad P_N = \mathbf{i}^T\mathbf{R}_s\,\mathbf{i}.$$

A.2. The test

Passivity test

The two-port is passive if and only if $\mathbf{R}_s$ is positive semidefinite. For a 2×2 matrix this means

$$r_{11} \geq 0, \qquad r_{22} \geq 0, \qquad r_{11}\, r_{22} - \tfrac{1}{4}\,(r_{12} + r_{21})^2 \geq 0.$$

If one condition fails, some port currents give $P_N < 0$: the two-port is active.

The same works with the other matrices. With $\mathbf{G}$, $P_N = \mathbf{v}^T\mathbf{G}\,\mathbf{v}$: test $\frac{1}{2}(\mathbf{G} + \mathbf{G}^T)$. With $\mathbf{H}$, take $\mathbf{x} = [\,i_1 \;\; v_2\,]^T$; then $[\,v_1 \;\; i_2\,]^T = \mathbf{H}\,\mathbf{x}$ and $P_N = \mathbf{x}^T\mathbf{H}\,\mathbf{x}$: test $\frac{1}{2}(\mathbf{H} + \mathbf{H}^T)$. The transmission matrix $\mathbf{T}$ does not give such a form, because it relates port 1 to port 2 instead of the voltages to the currents.

Example A.1 — the two-port of Example 14.4

$\mathbf{R} = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}$. It contains a dependent source. Is it passive?

Solution

Step 1. The symmetric part

$$\mathbf{R}_s = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

Step 2. The test

$3 \geq 0$, $1 \geq 0$ and $3 \cdot 1 - 0^2 = 3 \geq 0$. The two-port is **passive**: a dependent source inside does not make a two-port active by itself.

Example A.2 — the ideal transformer

$\mathbf{H} = \begin{bmatrix} 0 & n \\ -n & 0 \end{bmatrix}$ (Example 14.5). Is it passive?

Solution

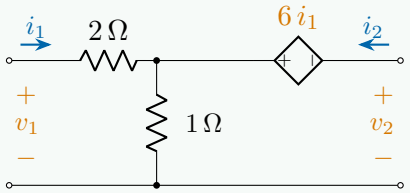
Step 1. The symmetric part

$\mathbf{H}$ is antisymmetric, $\mathbf{H}^T = -\mathbf{H}$, so $\frac{1}{2}(\mathbf{H} + \mathbf{H}^T) = \mathbf{0}$.

Step 2. The test

$P_N = 0$ for all port variables: the transformer is **passive** and absorbs no power, as in Example 14.8.

Example A.3 — a larger gain



The circuit of Example 14.2 with the dependent source $6\,i_1$ instead of $2\,i_1$. Is it passive?

Solution

Step 1. The matrix

As in Example 14.2, with 2 replaced by 6:

$$\mathbf{R} = \begin{bmatrix} 3 & 1 \\ 1-6 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ -5 & 1 \end{bmatrix}, \qquad \mathbf{R}_s = \begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix}.$$

Step 2. The test

$3 \cdot 1 - (-2)^2 = -1 < 0$: the two-port is **active**.

Step 3. A check

Take $i_1 = 1\,\text{A}$ and $i_2 = 2\,\text{A}$:

$$P_N = 3 \cdot 1^2 + (1-5) \cdot 1 \cdot 2 + 1 \cdot 2^2 = -1\,\text{W}.$$

The two-port delivers 1 W to the outside. ✓